

BEC303 Network Analysis — Module 1 Quiz

Basic Concepts — Mesh and Node Analysis

Field	Value
Subject code	BEC303
Module	1 — Basic Concepts — Mesh and Node Analysis
Duration	45 minutes
Maximum marks	50 (Part A 20 + Part B 30)
Attempts allowed	1
Passing marks	20
Textbook	W. H. Hayt, J. E. Kemmerly and S. M. Durbin, Engineering Circuit Analysis, 8th ed., 2012

Instructions

- Part A carries 20 one-mark multiple choice questions. Exactly one option is correct.
- Part B carries 6 descriptive questions totalling 30 marks.
- Non-programmable scientific calculators are permitted.
- All phase angles are to be reported in degrees, all currents in the direction marked.
- Show the equation you used; a bare numeric answer earns half the marks for that step.

Course outcome and Bloom coverage

Band	Questions	Share	Target
L1–L2 Remember / Understand	17	52%	~30%
L3–L4 Apply / Analyse	16	48%	~45%
L5–L6 Evaluate / Create	0	0%	~25%

Course outcome spread: CO1: 33. Total questions: 33.

Part A — Multiple choice (20 × 1 mark = 20 marks)

A1. Kirchhoff's Current Law is a direct consequence of which physical principle? [C01 | L1 | 1 mark | easy | Ch.3, pp.40-42]

- (A) Conservation of charge at a point that stores none
- (B) Conservation of energy around a loop
- (C) Ohm's law applied at a junction
- (D) Linearity of the circuit elements

A2. For a planar network with B branches and N nodes, how many independent KVL equations exist? [C01 | L2 | 1 mark | medium | Ch.3, pp.42-46]

- (A) $B - N + 1$

- (B) $N - 1$
- (C) $B - N$
- (D) $B + N - 1$

A3. A supernode is required when a voltage source is connected [C01 | L2 | 1 mark | easy | Ch.4, pp.89-92]

- (A) between two non-reference nodes
- (B) between a node and the reference node
- (C) in series with a resistor anywhere
- (D) in parallel with a current source

A4. In mesh analysis the physical current in a branch shared by two meshes is [C01 | L2 | 1 mark | medium | Ch.4, pp.92-98]

- (A) the difference of the two mesh currents
- (B) the sum of the two mesh currents
- (C) the larger of the two mesh currents
- (D) either mesh current, since they are equal there

A5. When a current source lies on a branch shared by two meshes, the correct response is to [C01 | L3 | 1 mark | medium | Ch.4, pp.98-101]

- (A) form a supermesh and write KVL around the merged outer path
- (B) write KVL through the source using its terminal voltage
- (C) replace the source by a short circuit
- (D) convert the source into a voltage source regardless of topology

A6. Which statement about source transformation is correct? [C01 | L2 | 1 mark | medium | Ch.5, pp.133-141]

- (A) It preserves terminal behaviour but not internal power dissipation
- (B) It preserves both terminal behaviour and internal dissipation
- (C) It may be applied to an ideal voltage source with no series resistance
- (D) It changes the load current by the ratio V_s/I_s

A7. A delta of $30\ \Omega$, $60\ \Omega$ and $30\ \Omega$ is converted to a wye. The branch at the terminal between the two $30\ \Omega$ arms is [C01 | L3 | 1 mark | hard | Ch.5, pp.154-157]

- (A) $7.5\ \Omega$
- (B) $15\ \Omega$
- (C) $30\ \Omega$
- (D) $60\ \Omega$

A8. Delta-wye conversion is specifically needed when [C01 | L2 | 1 mark | medium | Ch.5, pp.154-157]

- (A) no two elements are in simple series or parallel
- (B) the circuit contains a dependent source
- (C) the circuit is driven by a sinusoidal source
- (D) there are more meshes than nodes

A9. Expressed as a cosine, $5\sin(377t + 30^\circ)$ equals [C01 | L2 | 1 mark | medium | Ch.10, pp.371-374]

- (A) $5\cos(377t - 60^\circ)$
- (B) $5\cos(377t + 120^\circ)$

- (C) $5 \cdot \cos(377t + 30^\circ)$
- (D) $5 \cdot \cos(377t - 30^\circ)$

A10. The phasor of a sinusoid deliberately discards which piece of information? [C01 | L2 | 1 mark | easy | Ch.10, pp.383-389]

- (A) The frequency
- (B) The amplitude
- (C) The phase
- (D) The polarity reference

A11. The impedance of a 50 mH inductor at $\omega = 2000$ rad/s is [C01 | L2 | 1 mark | easy | Ch.10, pp.389-394]

- (A) $j100 \Omega$
- (B) $j0.1 \Omega$
- (C) $-j100 \Omega$
- (D) 100Ω

A12. A 100Ω resistor in series with $j100 \Omega$ has a total impedance magnitude of [C01 | L3 | 1 mark | medium | Ch.10, pp.389-394]

- (A) 141.4Ω
- (B) 200Ω
- (C) 100Ω
- (D) 0Ω

A13. Adding the phasors $30 \angle 30^\circ$ and $40 \angle -60^\circ$ gives [C01 | L3 | 1 mark | hard | Ch.10, pp.383-389]

- (A) $50 \angle -23.13^\circ$
- (B) $70 \angle -15^\circ$
- (C) $50 \angle 23.13^\circ$
- (D) $70 \angle -30^\circ$

A14. When applying superposition or finding a Thévenin resistance, dependent sources must be [C01 | L2 | 1 mark | medium | Ch.4, pp.80-101]

- (A) left active
- (B) set to zero
- (C) replaced by their average value
- (D) converted to independent sources

A15. An ideal independent voltage source is deactivated by replacing it with [C01 | L1 | 1 mark | easy | Ch.5, pp.123-133]

- (A) a short circuit
- (B) an open circuit
- (C) a resistor equal to its internal resistance
- (D) a current source of the same magnitude

A16. In nodal analysis the current through a resistor R between nodes v_1 and v_2 is written as [C01 | L2 | 1 mark | easy | Ch.4, pp.80-89]

- (A) $(v_1 - v_2)/R$
- (B) v_1/R
- (C) $(v_1 + v_2)/R$

- (D) v_2/R

A17. Mesh analysis requires the circuit to be planar because [C01 | L2 | 1 mark | medium | Ch.4, pp.92-98]

- (A) only a planar drawing has a well-defined complete set of independent meshes
- (B) complex numbers cannot be used in non-planar circuits
- (C) KVL fails in non-planar circuits
- (D) non-planar circuits have no nodes

A18. A complex forcing function is used because [C01 | L3 | 1 mark | hard | Ch.10, pp.374-383]

- (A) it solves a real and a quadrature problem at once, and the real part is kept
- (B) it approximates the sinusoid to within a few percent
- (C) it removes the transient response from the equation
- (D) it converts a nonlinear circuit into a linear one

A19. The reference node in nodal analysis is chosen [C01 | L2 | 1 mark | easy | Ch.4, pp.80-89]

- (A) freely, usually at the node with the most connections
- (B) only at a node physically connected to earth
- (C) at the node with the highest potential
- (D) at the node carrying the largest current

A20. A series RL circuit with $R = 3\ \Omega$ and $j\omega L = j4\ \Omega$ driven by $10\angle 0^\circ\text{ V}$ carries a current of [C01 | L4 | 1 mark | hard | Ch.10, pp.374-394]

- (A) $2\angle -53.13^\circ\text{ A}$
- (B) $2\angle +53.13^\circ\text{ A}$
- (C) $1.43\angle 0^\circ\text{ A}$
- (D) $10\angle -53.13^\circ\text{ A}$

Part B — Descriptive (13 questions, 30 marks)

B1. State Kirchhoff's Current Law and explain why it may be applied to a closed boundary enclosing several nodes. [C01 | L2 | 3 marks | Ch.3, pp.40-42]

B2. A 6 mA source feeds node v_1 ; 2 k Ω runs v_1 to ground, 1 k Ω runs v_1 to v_2 , and 3 k Ω runs v_2 to ground. Find v_1 and v_2 by nodal analysis. [C01 | L3 | 3 marks | Ch.4, pp.80-89]

B3. Explain when a supernode is required, and state the two equations it produces. [C01 | L3 | 3 marks | Ch.4, pp.89-92]

B4. Two meshes share a 3 mA current source. Mesh 1 contains a 12 V source and 2 k Ω ; mesh 2 contains 4 k Ω . Find both mesh currents. [C01 | L3 | 3 marks | Ch.4, pp.98-101]

B5. Convert a delta of 30 Ω , 60 Ω and 30 Ω to its equivalent wye. [C01 | L3 | 2 marks | Ch.5, pp.154-157]

B6. A 100 Ω resistor is in series with $j100\ \Omega$, driven by $10\angle 0^\circ\text{ V}$. Find the current and explain why the two element voltages do not add arithmetically to 10 V. [C01 | L4 | 2 marks | Ch.10, pp.389-394]

B7. Distinguish an independent source from a dependent source, and state how each is treated when sources are deactivated. [C01 | L2 | 2 marks | Ch.2, pp.17-22]

B8. For a planar network with B branches and N nodes, state how many independent KCL and KVL equations exist, and justify both counts. [C01 | L3 | 2 marks | Ch.3, pp.39-46]

- B9.** A 12 V source in series with 4 k Ω feeds a 2 k Ω load. Solve for the load voltage using a source transformation, and state one quantity the transformed circuit cannot be used to find. [C01 | L3 | 2 marks | Ch.5, pp.133-141]
- B10.** Express $5\sin(377t + 30^\circ)$ as a cosine and compare its phase with $4\cos(377t - 20^\circ)$. [C01 | L2 | 2 marks | Ch.10, pp.371-374]
- B11.** Add the phasors $30\angle 30^\circ$ and $40\angle -60^\circ$, showing the rectangular intermediate step. [C01 | L3 | 2 marks | Ch.10, pp.383-389]
- B12.** Explain what information a phasor discards and why discarding it is safe. [C01 | L4 | 2 marks | Ch.10, pp.383-389]
- B13.** A network contains a VCCS whose control voltage is a node voltage. Describe how nodal analysis proceeds and what changes in the coefficient matrix. [C01 | L4 | 2 marks | Ch.4, pp.80-101]

Answer key — Part A

Q	Answer	CO	Bloom	Reference
A1	A	CO1	L1	Ch.3, pp.40-42
A2	A	CO1	L2	Ch.3, pp.42-46
A3	A	CO1	L2	Ch.4, pp.89-92
A4	A	CO1	L2	Ch.4, pp.92-98
A5	A	CO1	L3	Ch.4, pp.98-101
A6	A	CO1	L2	Ch.5, pp.133-141
A7	A	CO1	L3	Ch.5, pp.154-157
A8	A	CO1	L2	Ch.5, pp.154-157
A9	A	CO1	L2	Ch.10, pp.371-374
A10	A	CO1	L2	Ch.10, pp.383-389
A11	A	CO1	L2	Ch.10, pp.389-394
A12	A	CO1	L3	Ch.10, pp.389-394
A13	A	CO1	L3	Ch.10, pp.383-389
A14	A	CO1	L2	Ch.4, pp.80-101
A15	A	CO1	L1	Ch.5, pp.123-133
A16	A	CO1	L2	Ch.4, pp.80-89
A17	A	CO1	L2	Ch.4, pp.92-98
A18	A	CO1	L3	Ch.10, pp.374-383
A19	A	CO1	L2	Ch.4, pp.80-89
A20	A	CO1	L4	Ch.10, pp.374-394

Part A — why each option is right or wrong

A1. Correct answer: (A) Conservation of charge at a point that stores none

- (A) Correct — a lumped node has no capacitance to store charge, so current in must equal current out at every instant.
- (B) Wrong - energy conservation around a loop gives KVL, not KCL.
- (C) Wrong - KCL holds even for nonlinear and non-ohmic branches, so it cannot rest on Ohm's law.
- (D) Wrong - KCL holds for nonlinear circuits too, so linearity is not required.

A2. Correct answer: (A) $B - N + 1$

- (A) Correct — $B - N + 1$ is the mesh count, and each mesh contributes exactly one independent loop equation.
- (B) Wrong - $N - 1$ is the number of independent KCL equations, not KVL.
- (C) Wrong - this undercounts by one; a single-loop circuit with $B = N$ would then need zero equations.
- (D) Wrong - this overcounts badly and would produce a dependent, unsolvable set.

A3. Correct answer: (A) between two non-reference nodes

- (A) Correct — with neither end at the reference, neither node voltage is known and the source branch has no current expression.
- (B) Wrong - a source with one end grounded fixes that node voltage directly and needs no supernode.
- (C) Wrong - a series resistor gives the branch a perfectly ordinary current expression.
- (D) Wrong - that topology affects source transformation, not the supernode condition.

A4. Correct answer: (A) the difference of the two mesh currents

- (A) Correct — the two circulating currents traverse the shared branch in opposite senses, so they subtract.
- (B) Wrong - they would only add if both meshes were assigned opposing rotation directions, which the standard convention forbids.
- (C) Wrong - this confuses magnitude with algebraic combination and gives no defensible sign.
- (D) Wrong - mesh currents are generally different; only their difference is physically measurable in that branch.

A5. Correct answer: (A) form a supermesh and write KVL around the merged outer path

- (A) Correct — the voltage across an ideal current source is unknown, so the traversal path must go around it.
- (B) Wrong - that terminal voltage is an unknown you cannot express, so the equation becomes unsolvable.
- (C) Wrong - shorting a current source changes the circuit; sources are only deactivated during superposition.
- (D) Wrong - source transformation needs a parallel resistance, which a shared ideal source does not have.

A6. Correct answer: (A) It preserves terminal behaviour but not internal power dissipation

- (A) Correct — the load cannot distinguish the two forms, but the resistance sits in a different position and dissipates differently.
- (B) Wrong - internal dissipation is exactly what is not preserved, which is why efficiency must be computed on the original.
- (C) Wrong - the transformation requires division by R , so an ideal source cannot be transformed.

- (D) Wrong - the load current is identical before and after; that is the definition of terminal equivalence.

A7. Correct answer: (A) $7.5\ \Omega$

- (A) Correct — $(30 \times 30)/120 = 7.5\ \Omega$, using the two arms that touch that terminal.
- (B) Wrong - 15 ohm is what the other two wye branches come to, each using one 30 ohm and the 60 ohm arm.
- (C) Wrong - this simply repeats a delta value and ignores the conversion formula entirely.
- (D) Wrong - this uses the arm opposite the terminal, which is the classic sign of applying the formula backwards.

A8. Correct answer: (A) no two elements are in simple series or parallel

- (A) Correct — the bridge topology stalls series-parallel reduction, and conversion is the move that unblocks it.
- (B) Wrong - dependent sources are handled by auxiliary equations; the conversion applies only to passive three-terminal networks.
- (C) Wrong - frequency is irrelevant to the topological deadlock the conversion solves.
- (D) Wrong - the mesh-to-node ratio guides which analysis method to pick, not whether a bridge exists.

A9. Correct answer: (A) $5\cos(377t - 60^\circ)$

- (A) Correct — $\sin(x) = \cos(x - 90^\circ)$, so $30^\circ - 90^\circ = -60^\circ$.
- (B) Wrong - this adds 90 deg instead of subtracting it, the most common sign slip in the conversion.
- (C) Wrong - this drops the conversion altogether and treats sine and cosine as interchangeable.
- (D) Wrong - this subtracts 60 deg from nothing; the shift is a fixed 90 deg, not the phase value itself.

A10. Correct answer: (A) The frequency

- (A) Correct — a linear circuit cannot change frequency, so carrying it through every equation is redundant.
- (B) Wrong - amplitude is the magnitude of the phasor and is fully retained.
- (C) Wrong - phase is the angle of the phasor and is precisely what the representation is for.
- (D) Wrong - the polarity reference must still be marked on the schematic for the phasor to be interpreted.

A11. Correct answer: (A) $j100\ \Omega$

- (A) Correct — $Z_L = j\omega L = j \times 2000 \times 0.05 = j100\ \Omega$.
- (B) Wrong - this divides by omega instead of multiplying, the capacitive formula applied to an inductor.
- (C) Wrong - the sign is wrong; negative reactance is capacitive, and an inductor is always +j.
- (D) Wrong - the magnitude is right but dropping the j loses the 90 deg phase lead entirely.

A12. Correct answer: (A) $141.4\ \Omega$

- (A) Correct — $|100 + j100| = \sqrt{(100^2 + 100^2)} = 141.4\ \Omega$ at an angle of 45° .
- (B) Wrong - this adds magnitudes arithmetically, a 41 percent overestimate that corrupts every later power figure.
- (C) Wrong - this ignores the reactive part completely.
- (D) Wrong - cancellation to zero requires equal and opposite reactances, not a resistance and a reactance.

A13. Correct answer: (A) $50\angle -23.13^\circ$

- (A) Correct — in rectangular form the sum is $45.98 - j19.64$, whose magnitude is 50 and angle -23.13° .

- (B) Wrong - this adds magnitudes and averages angles, which is never valid for phasor addition.
- (C) Wrong - the magnitude is right but the sign of the angle is inverted; the imaginary part is negative here.
- (D) Wrong - both the magnitude and angle come from treating phasors as ordinary scalars.

A14. Correct answer: (A) left active

- (A) Correct — a dependent source expresses the circuit's internal structure, not an external excitation.
- (B) Wrong - zeroing it changes the network into a different circuit and gives a wrong resistance.
- (C) Wrong - a dependent source has no meaningful average value independent of the circuit state.
- (D) Wrong - its value tracks another circuit variable, so it cannot be frozen into an independent value.

A15. Correct answer: (A) a short circuit

- (A) Correct — zero volts across a branch is exactly a short circuit.
- (B) Wrong - an open circuit deactivates a current source, not a voltage source.
- (C) Wrong - an ideal source has no internal resistance to substitute.
- (D) Wrong - substituting another live source does not deactivate anything.

A16. Correct answer: (A) $(v_1 - v_2)/R$

- (A) Correct — the branch current depends on the difference in potential across the element, never on one end alone.
- (B) Wrong - this silently grounds the far end of a resistor that is not grounded.
- (C) Wrong - potentials subtract across an element; adding them has no physical meaning.
- (D) Wrong - same error as option B, with the wrong end grounded instead.

A17. Correct answer: (A) only a planar drawing has a well-defined complete set of independent meshes

- (A) Correct — a mesh is a loop enclosing no other loop, which is a property of a crossing-free drawing.
- (B) Wrong - complex arithmetic is indifferent to topology.
- (C) Wrong - KVL holds for any closed path in any lumped circuit, planar or not.
- (D) Wrong - every circuit has nodes; planarity is about how loops nest, not about node existence.

A18. Correct answer: (A) it solves a real and a quadrature problem at once, and the real part is kept

- (A) Correct — superposition guarantees that the real part of the response belongs to the real part of the drive.
- (B) Wrong - the method is exact, not approximate; no accuracy is traded away.
- (C) Wrong - the transient is discarded by restricting attention to steady state, not by the complex form itself.
- (D) Wrong - the method requires linearity to begin with and cannot create it.

A19. Correct answer: (A) freely, usually at the node with the most connections

- (A) Correct — the choice removes an arbitrary additive constant and is made for convenience alone.
- (B) Wrong - the analysis reference has nothing to do with a physical earth connection.
- (C) Wrong - the reference is assigned zero volts by definition, so it cannot be chosen by potential.
- (D) Wrong - branch currents are outputs of the analysis and are unknown when the reference is chosen.

A20. Correct answer: (A) $2\angle -53.13^\circ$ A

- (A) Correct — $I = 10\angle 0^\circ / (3 + j4) = 10\angle 0^\circ / 5\angle 53.13^\circ = 2\angle -53.13^\circ$ A.
- (B) Wrong - the sign is inverted; current lags voltage in an inductive circuit, so the angle must be negative.

- (C) Wrong - this divides by $3 + 4 = 7$ as if the impedance were real.
- (D) Wrong - the angle is right but the magnitude was never divided by the impedance magnitude of 5.

Part B — model answers and marking scheme

B1 (3 marks) — CO1, L2, Ch.3, pp.40-42

Question. State Kirchhoff's Current Law and explain why it may be applied to a closed boundary enclosing several nodes.

Model answer. KCL states that the algebraic sum of currents entering any node is zero at every instant, because a lumped node stores no charge. The same argument applies to a closed boundary drawn around a group of nodes: charge cannot accumulate anywhere inside, so the net current crossing the boundary must be zero. Branches lying wholly inside contribute equally to entering and leaving and therefore cancel. This generalisation is what licenses the supernode construction.

Point required	Marks
Correct statement of KCL with the instantaneous qualifier	1
Charge-conservation justification and the boundary generalisation	2
Total	3

B2 (3 marks) — CO1, L3, Ch.4, pp.80-89

Question. A 6 mA source feeds node v1; 2 k Ω runs v1 to ground, 1 k Ω runs v1 to v2, and 3 k Ω runs v2 to ground. Find v1 and v2 by nodal analysis.

Model answer. KCL at v1: $v_1/2000 + (v_1 - v_2)/1000 = 0.006$. KCL at v2: $(v_2 - v_1)/1000 + v_2/3000 = 0$. The second gives $v_1 = (4/3)v_2$. Substituting: $(4/3)v_2/2000 + (1/3)v_2/1000 = 0.006$, so $v_2 = 6$ V and $v_1 = 8$ V. Check: the 1 k Ω carries $(8 - 6)/1000 = 2$ mA, matching the 6 V/3 k $\Omega = 2$ mA leaving v2.

Point required	Marks
Both KCL equations with correct difference terms	2
Correct solution plus a consistency check	1
Total	3

B3 (3 marks) — CO1, L3, Ch.4, pp.89-92

Question. Explain when a supernode is required, and state the two equations it produces.

Model answer. A supernode is required when an ideal voltage source connects two non-reference nodes, so neither node voltage is known and the source branch admits no current expression. Enclose both nodes and the source in a boundary and apply KCL to it; the source current is internal and cancels. The second equation is the source constraint $v_a - v_b = V_s$. Together these replace the two lost node equations, so the equation count is unchanged.

Point required	Marks
Condition under which a supernode is needed	1
Boundary KCL and the constraint equation, with the count argument	2

Point required	Marks
Total	3

B4 (3 marks) — CO1, L3, Ch.4, pp.98-101

Question. Two meshes share a 3 mA current source. Mesh 1 contains a 12 V source and 2 k Ω ; mesh 2 contains 4 k Ω . Find both mesh currents.

Model answer. The current source lies on the shared branch so its voltage is unknown and a supermesh is formed. Constraint: $i_1 - i_2 = 3$ mA. Supermesh KVL round the outer path: $-12 + 2000 \cdot i_1 + 4000 \cdot i_2 = 0$. Substituting $i_1 = i_2 + 0.003$ gives $6000 \cdot i_2 = 6$, so $i_2 = 1$ mA and $i_1 = 4$ mA.

Point required	Marks
Correct constraint and recognition that a supermesh is needed	2
Correct supermesh KVL and numerical solution	1
Total	3

B5 (2 marks) — CO1, L3, Ch.5, pp.154-157

Question. Convert a delta of 30 Ω , 60 Ω and 30 Ω to its equivalent wye.

Model answer. Sum of the delta arms = 120 Ω . Each wye branch is the product of the two delta arms meeting at that terminal divided by the sum. The terminal between the two 30 Ω arms gives $(30 \times 30)/120 = 7.5$ Ω ; the other two give $(30 \times 60)/120 = 15$ Ω each. The equivalence holds at the three terminals only.

Point required	Marks
Correct adjacent-arms formula	1
All three values correct	1
Total	2

B6 (2 marks) — CO1, L4, Ch.10, pp.389-394

Question. A 100 Ω resistor is in series with j100 Ω , driven by $10\angle 0^\circ$ V. Find the current and explain why the two element voltages do not add arithmetically to 10 V.

Model answer. $Z = 100 + j100 = 141.4\angle 45^\circ$ Ω , so $I = 10\angle 0^\circ / 141.4\angle 45^\circ = 70.7\angle -45^\circ$ mA. $V_R = 7.07\angle -45^\circ$ V and $V_L = 7.07\angle 45^\circ$ V. Their magnitudes sum to 14.14 V, but they are 90° apart, so the phasor sum is $\sqrt{(7.07^2 + 7.07^2)} = 10$ V. KVL holds for phasors, not for magnitudes.

Point required	Marks
Correct impedance and current	1
Correct element voltages and the phasor-sum explanation	1
Total	2

B7 (2 marks) — CO1, L2, Ch.2, pp.17-22

Question. Distinguish an independent source from a dependent source, and state how each is treated when sources are deactivated.

Model answer. An independent source fixes its voltage or current at a value determined outside the circuit. A dependent source fixes its output as a multiple of another voltage or current inside the same circuit, and is drawn as a diamond. When sources are deactivated for superposition or a look-back resistance, independent voltage sources become shorts and independent current sources become opens. Dependent sources are never deactivated, because they express circuit structure rather than excitation.

Point required	Marks
Correct distinction with the controlling-variable idea	1
Correct deactivation rules including the dependent-source exception	1
Total	2

B8 (2 marks) — CO1, L3, Ch.3, pp.39-46

Question. For a planar network with B branches and N nodes, state how many independent KCL and KVL equations exist, and justify both counts.

Model answer. There are $N - 1$ independent KCL equations: one per non-reference node, since the Nth is the negated sum of the others and adds nothing. There are $B - N + 1$ independent KVL equations, which is the number of meshes; any larger loop equation is a linear combination of mesh equations. Together the two sets give B equations for B unknown branch currents.

Point required	Marks
Both counts stated correctly	1
Correct justification of the redundancy in each case	1
Total	2

B9 (2 marks) — CO1, L3, Ch.5, pp.133-141

Question. A 12 V source in series with 4 k Ω feeds a 2 k Ω load. Solve for the load voltage using a source transformation, and state one quantity the transformed circuit cannot be used to find.

Model answer. Transform to $12/4000 = 3$ mA in parallel with 4 k Ω . The 4 k Ω and 2 k Ω are now in parallel, giving 1.333 k Ω , so $V_{\text{load}} = 3 \text{ mA} \times 1.333 \text{ k}\Omega = 4 \text{ V}$. This agrees with the direct divider $12 \times 2/6 = 4 \text{ V}$. The transformed circuit cannot be used to find the power dissipated inside the original source, because only terminal behaviour is preserved.

Point required	Marks
Correct transformation and load voltage	1
Correct statement of the internal-power limitation	1
Total	2

B10 (2 marks) — CO1, L2, Ch.10, pp.371-374

Question. Express $5 \cdot \sin(377t + 30^\circ)$ as a cosine and compare its phase with $4 \cdot \cos(377t - 20^\circ)$.

Model answer. Using $\sin(x) = \cos(x - 90^\circ)$, $5 \cdot \sin(377t + 30^\circ) = 5 \cdot \cos(377t - 60^\circ)$. Both are now cosines at $\omega = 377 \text{ rad/s}$. The phase difference is $-60^\circ - (-20^\circ) = -40^\circ$, so the first signal lags the second by 40° . Comparing the written angles 30° and -20° without converting would have given a spurious 50° lead.

Point required	Marks
Correct conversion to cosine form	1
Correct phase difference with lead/lag stated	1
Total	2

B11 (2 marks) — CO1, L3, Ch.10, pp.383-389

Question. Add the phasors $30\angle 30^\circ$ and $40\angle -60^\circ$, showing the rectangular intermediate step.

Model answer. $30\angle 30^\circ = 25.98 + j15.00$ and $40\angle -60^\circ = 20.00 - j34.64$. Adding rectangular components gives $45.98 - j19.64$. The magnitude is $\sqrt{(45.98^2 + 19.64^2)} = 50.0$ and the angle is $\arctan(-19.64/45.98) = -23.13^\circ$. So the sum is $50\angle -23.13^\circ$. Addition is valid only in rectangular form.

Point required	Marks
Both phasors correctly converted to rectangular form	1
Correct sum converted back to polar form	1
Total	2

B12 (2 marks) — CO1, L4, Ch.10, pp.383-389

Question. Explain what information a phasor discards and why discarding it is safe.

Model answer. A phasor retains amplitude and phase and discards frequency. This is safe because a linear circuit cannot change frequency: every branch response to a sinusoidal excitation is a sinusoid at the same ω , so carrying ω through every equation is redundant. The price is that a phasor describes steady state only and says nothing about the transient, and that phasors at different frequencies cannot be added.

Point required	Marks
Correct identification of what is discarded and why it is redundant	1
Correct statement of the steady-state limitation	1
Total	2

B13 (2 marks) — CO1, L4, Ch.4, pp.80-101

Question. A network contains a VCCS whose control voltage is a node voltage. Describe how nodal analysis proceeds and what changes in the coefficient matrix.

Model answer. The nodal equations are written exactly as if all sources were independent, with the dependent source appearing as a current term. Because the controlling variable is already a chosen node voltage, it is substituted directly and no auxiliary equation is needed. The number of unknowns is unchanged. The coefficient matrix, however, loses its symmetry: the dependent source adds a term to one row with no matching term in the mirrored position.

Point required	Marks
Correct procedure including the substitution of the control	1
Correct statement about the loss of symmetry	1
Total	2